

Kuan-Hung (Peter) Yeh

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Research Interest

My research focuses on statistical methodology and application to **causal inference** to generate regulatory-grade real-world evidence. My work spans both clinical and statistical research, evaluating **treatment safety** and **effectiveness** in gastroenterology, while developing robust methods for the joint design and analysis of complex **experimental** and **observational studies**. I am especially interested in how **real-world evidence** inform evidence-based decisions across drug discovery, clinical development, regulatory, and commercialization.

Education

University of California, San Diego <i>Ph.D. in Biomedical Informatics (Advisors: Dr. Ronghui Xu and Siddharth Singh)</i>	San Diego, CA 2023 – 03/2028 (Exp)
University of California, Los Angeles <i>Master of Science in Biostatistics (Advisor: Dr. Bogdan Pasaniuc)</i>	Los Angeles, CA 2021 – 2023
National Taiwan University <i>Bachelor of Science in Public Health, Biostatistics Track</i>	Taipei, Taiwan 2016 – 2020

Fellowships

FDA-OCE-ASA Oncology Educational Fellow <i>FDA Oncology Center of Excellence & American Statistical Association (BIOP Section)</i>	Oct. 2026 – May 2027 <i>Virtual</i>
Selected as 1 of 20 fellows nationwide for a highly selective FDA–American Statistical Association fellowship focused on oncology drug development and regulatory policy.	

Technical Skills

Methods: Biostatistics, causal inference, target trial emulation, propensity score methods, survival analysis, multiple imputation, resampling & permutation-based inference, distributional/covariate shift
Languages: R (tidyverse, survival, MatchIt, mice), Python (pandas, NumPy, scikit-learn, statsmodels), SQL, LaTeX
Tools & Computing: Git, HTCondor/HPC job submission, R Shiny

Experience

Graduate Researcher <i>Department of Biomedical Informatics, UCSD Health</i>	Jan. 2024 – Present <i>La Jolla, CA</i>
- Generated real-world evidence in Gastroenterology using Optum claims and Epic Cosmos datasets, enhancing treatment strategy evaluation through target trial emulation, survival analysis, and propensity score methods. - Developed and implemented a data fusion framework for ATE estimation under distributional shift, comparing resampling to weighting and extending to generate indirect comparisons across clinical trials Collaborated with multiple clinical teams to design and conduct research studies, communicating scientific findings effectively and proactively solving problems to inform treatment safety and efficacy. - Contributed to the design and physician evaluation of a multi-agent LLM framework for ICU readmission risk stratification and discharge planning (manuscript under review)	
Biostatistician Intern <i>Foundation Medicine, Inc.</i>	Jun. 2022 – Sep. 2022 <i>Boston, MA</i>
- Contributed to the development of statistical software tools, including R function and Shiny Apps [Demo]. - Developed new criteria using simulation for diagnostic assay precision studies, boosting statistical power by 90%. - Conducted statistical analyses (e.g., sample size, power, precision) to ensure quality assurance for new reagent design in FoundationOne Liquid CDx.	
Graduate Consultant <i>Department of Computational Medicine & Medicine Statistics, UCLA</i>	May. 2022 – Jun. 2023 <i>Los Angeles, CA</i>
- Performed mediation analyses and data visualizations for a three-arm randomized controlled trial. - Developed a survival data simulation pipeline to estimate how genetic architecture impacts in survival modeling and polygenic risk score stratification.	

Updated September 26, 2026

Honors & Awards

1. FDA-OCE-ASA Oncology Educational Fellow, FDA OCE & ASA, 2026
2. Travel Award, IMSI Workshop, National Science Foundation (NSF), Chicago, 2026.
3. Best Poster Presentation Award, 16th GI/Hepatology Annual Research Symposium, UCSD Health, 2025.
4. J. Yang Scholarship, Department of Bioinformatics, University of California, San Diego, 2023.
5. Honor Graduate, Department of Public Health, National Taiwan University, 2020.
6. Dr. K. P. Chen Memorial Scholarship, Department of Public Health, National Taiwan University, 2019.

Publications

† co-first author * corresponding author

Peer-Reviewed Journal Articles

1. **Yeh, K. H.**, Ahuja, D., Patel, S. B., Xu, R., Park, S., . . . , Singh, S.* (2026). Adjunctive GLP-1 Receptor Agonists in Patients with Inflammatory Bowel Diseases and Obesity and/or Diabetes: A Target Trial Emulation. *Clinical Gastroenterology and Hepatology*.
2. Ahuja, D., Park, S. K., **Yeh, K. H.**†, Patel, S. B., Goodwin, S. W., . . . , Singh, S.* (2026). Overall and Comparative Risk of Shingles with Advanced Therapies in Patients with Inflammatory Bowel Diseases. *Alimentary Pharmacology & Therapeutics*, 63(5), 686–695.
3. Park, S., Ahuja, D., **Yeh, K. H.**†, Patel, S. B., Goodwin, S. W., . . . , Singh, S.* (2026). Comparative Safety of Advanced Therapies for Crohn Disease. *JAMA Network Open*, 9(2), e2557922.
4. Ahuja, D., **Yeh, K. H.**†, Patel, S. B., Park, S. K., Xu, R., & Singh, S.* (2025). Risk of Tuberculosis with Advanced Therapies in 20,705 Patients with Inflammatory Bowel Diseases in Low Incidence Regions—A US Claims-Based Study. *Alimentary Pharmacology & Therapeutics*.
5. Ahuja, D., Dziegielewska, C., **Yeh, K. H.**†, Patel, S. B., Goodwin, S. W., . . . , Singh, S.* (2025). Comparative Safety of Advanced Therapies in Patients with Ulcerative Colitis: An Administrative Claims-Based Study. *American Journal of Gastroenterology*.
6. Corroon, J., Bradley, R., Grant, I., Bancks, M. P., . . . **Yeh, K. H.**, et al. (2025). Lifetime Cannabis Use and Incident Hypertension: The Coronary Artery Risk Development in Young Adults (CARDIA) Study. *Hypertension*, 82(10), 1641–1652.
7. Solitano, V., Ahuja, D., Lee, H. H., Gaikwad, R., **Yeh, K. H.**, . . . , Singh, S.* (2025). Comparative Safety of JAK Inhibitors vs TNF Antagonists in Immune-Mediated Inflammatory Diseases: A Systematic Review and Meta-Analysis. *JAMA Network Open*, 8(9), e2531204.
8. Ahuja, D., **Yeh, K. H.**†, Patel, S. B., Lee, H. H., Goodwin, S. W., . . . , Singh, S.* (2025). Comparative Safety of Janus Kinase Inhibitors vs. Tumor Necrosis Factor Antagonists in Patients with Inflammatory Bowel Diseases. *Clinical Gastroenterology and Hepatology*.
9. McCurdy, J. D.*, Munir, J., Parlow, S., Sambhi, G., . . . **Yeh, K. H.**, et al. (2025). The Impact of Setons on Perianal Fistula Outcomes in Patients with Crohn’s Disease Treated with Anti-TNF Therapy: A Multicentre Study. *Alimentary Pharmacology & Therapeutics*, 61(10), 1671–1679.
10. Haynesworth, A., **Yeh, K. H.**†, Lee, H. H., Kirkpatrick, M., . . . , Singh, S.* (2025). Long-Term Outcomes of an Infliximab-First Versus Vedolizumab-First Treatment Strategy in Biologic-Naïve Patients with Ulcerative Colitis. *Alimentary Pharmacology & Therapeutics*, 61(4), 693–701.
11. Chan, H. C., Huang, C. C., Huang, C. C., Chattopadhyay, A., **Yeh, K. H.**, . . . , Lu, T. P.* (2022). Predicting Colon Cancer-Specific Survival for the Asian Population Using National Cancer Registry Data from Taiwan. *Annals of Surgical Oncology*, 29(2), 853–863.

Peer-Reviewed Conference Proceedings

1. **Yeh, K. H.**, Lin, C. H., Hsiao, T. H., & Lu, T. P.* (2020). Genome-Wide Association Study (GWAS) on Metabolic Syndrome in Subjects with Abdominal Obesity in a Taiwanese Population. *2020 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*.

Selected Conference Abstracts & Presentations

1. **Yeh, K. H.**, Xu, R., & Singh, S.* (2026). Trial Data Fusion: Indirect Comparisons Beyond Covariate Shift. Presented at the American Causal Inference Conference (ACIC) 2026, Poster Presentation. (*Presenting Author*)
2. Ahuja, D., **Yeh, K. H.**, Park, S. K., Qi, Y., Patel, S., . . . , Singh, S.* (2026). Tu1584: Cancer Risk with Advanced Therapies in Patients with Inflammatory Bowel Diseases: An Administrative Claims-Based Study. Presented at Digestive Disease Week (DDW) 2026. *Gastroenterology*, 170(6), S-1314.
3. Hupé, M., Ahuja, D., **Yeh, K. H.**, Patel, S., Xu, R., . . . , Singh, S.* (2026). Mo1554: Comparative Effectiveness of Advanced Therapies for Preventing Penetrating Complications in Crohn's Disease: An Administrative Claims-Based Study. Presented at Digestive Disease Week (DDW) 2026. *Gastrointestinal Endoscopy*, 103(5), S-1185.
4. Ahuja, D., Park, S. K., **Yeh, K. H.**, Patel, S., Goodwin, S., . . . , Singh, S.* (2025). S1872: Overall and Comparative Risk of Shingles with Advanced Therapies in Patients with Inflammatory Bowel Diseases. Presented at the ACG Annual Scientific Meeting 2025. *American Journal of Gastroenterology*, 120(10S2).
5. Lee, H. H., Ahuja, D., Solitano, V., Gaikwad, R., **Yeh, K. H.**, . . . , Singh, S.* (2025). Tu1850: Comparative Risk of Major Adverse Cardiovascular Events with JAK Inhibitors vs. TNF Antagonists in Patients with Immune-Mediated Inflammatory Disorders: A Systematic Review and Meta-Analysis. Presented at Digestive Disease Week (DDW) 2025. *Gastroenterology*, 169(1), S-1522.
6. Gaikwad, R., Ahuja, D., Solitano, V., Lee, H. H., **Yeh, K. H.**, . . . , Jairath, V.* (2025). Tu1849: Comparative Risk of Venous Thromboembolism with JAK Inhibitors vs. TNF Antagonists in Patients with Immune-Mediated Inflammatory Disorders: A Systematic Review and Meta-Analysis. Presented at Digestive Disease Week (DDW) 2025. *Gastroenterology*, 169(1), S-1522.
7. McCurdy, J. D.*, Munir, J., Parlow, S., Sambhi, G., . . . **Yeh, K. H.**, et al. (2025). 124: The Impact of Setons on Perianal Fistula Outcomes in Patients with Crohn's Disease Treated with Anti-TNF Therapy: A Multicenter Study. Presented at Digestive Disease Week (DDW) 2025. *Gastroenterology*, 169(1), S-35–S-36.
8. Pandita, P., Lee, H. H., **Yeh, K. H.**, Haynesworth, A., . . . , Singh, S.* (2025). Tu1856: Treatment Sequencing: Long-Term Outcomes of an Infliximab-First vs. Ustekinumab-First Treatment Strategy in Biologic-Naive Patients with Crohn's Disease. Presented at Digestive Disease Week (DDW) 2025. *Gastroenterology*, 169(1), S-1525.

Manuscripts Under Review & In Preparation

1. Lee, H., . . . **Yeh, K. H.**, et al. (2026). "Comparative Long-term Outcomes of Treatment Strategies in Biologic-Naive Patients with Crohn's Disease: Anti-TNF-first vs. Ustekinumab-first" *Alimentary Pharmacology & Therapeutics* (under revision).
2. Hupé, M., . . . **Yeh, K. H.**, et al. (2026). "Risk of Perianal Penetrating Complications Among Patients with Non-Penetrating Crohn's Disease Initiating Advanced Therapies: A Retrospective Cohort Study." *Journal of Crohn's and Colitis* (under revision).
3. Silverman, A.*, . . . **Yeh, K. H.**, et al. (2026). "Combining Immunomodulators with Biologic Therapies vs. Biologic Monotherapy in Patients with Inflammatory Bowel Diseases: Target Trial Emulations." *Therapeutic Advances in Gastroenterology* (under revision).
4. Ahuja, D., . . . **Yeh, K. H.**, . . . , Singh, S.* (2026). "Upadacitinib Is Associated with the Lowest Risk of Treatment Failure Among Advanced Therapies for Ulcerative Colitis." *Clinical Gastroenterology and Hepatology* (under revision).
5. Ahuja, D., **Yeh, K. H.**, Park, S. K., Qi, Y., Patel, S., . . . , Singh, S.* (2026). "Cancer Risk with Advanced Therapies in Patients with Inflammatory Bowel Diseases: An Administrative Claims-Based Study." *Clinical Gastroenterology and Hepatology* (under revision).
6. Neel, N., . . . **Yeh, K. H.**, et al. (2026). "Physician Evaluation of a Multi-Agent LLM Framework for ICU Readmission Risk Stratification and Discharge Planning." Submitted (under review).
7. **Yeh, K. H.**, Sitapati A., Singh, S., & Xu, R.* (2026). "Trial Data Fusion: Indirect Comparisons Beyond Covariate Shift." *Manuscript in preparation*.
8. Singh, S., **Yeh, K. H.**, Silverman, A. L., Ahuja, D., . . . & Xu, R. (2026). "Comparative Effectiveness Research in Gastroenterology and Hepatology: A Practical Guide to Causal Inference Methods." *Manuscript in preparation*.